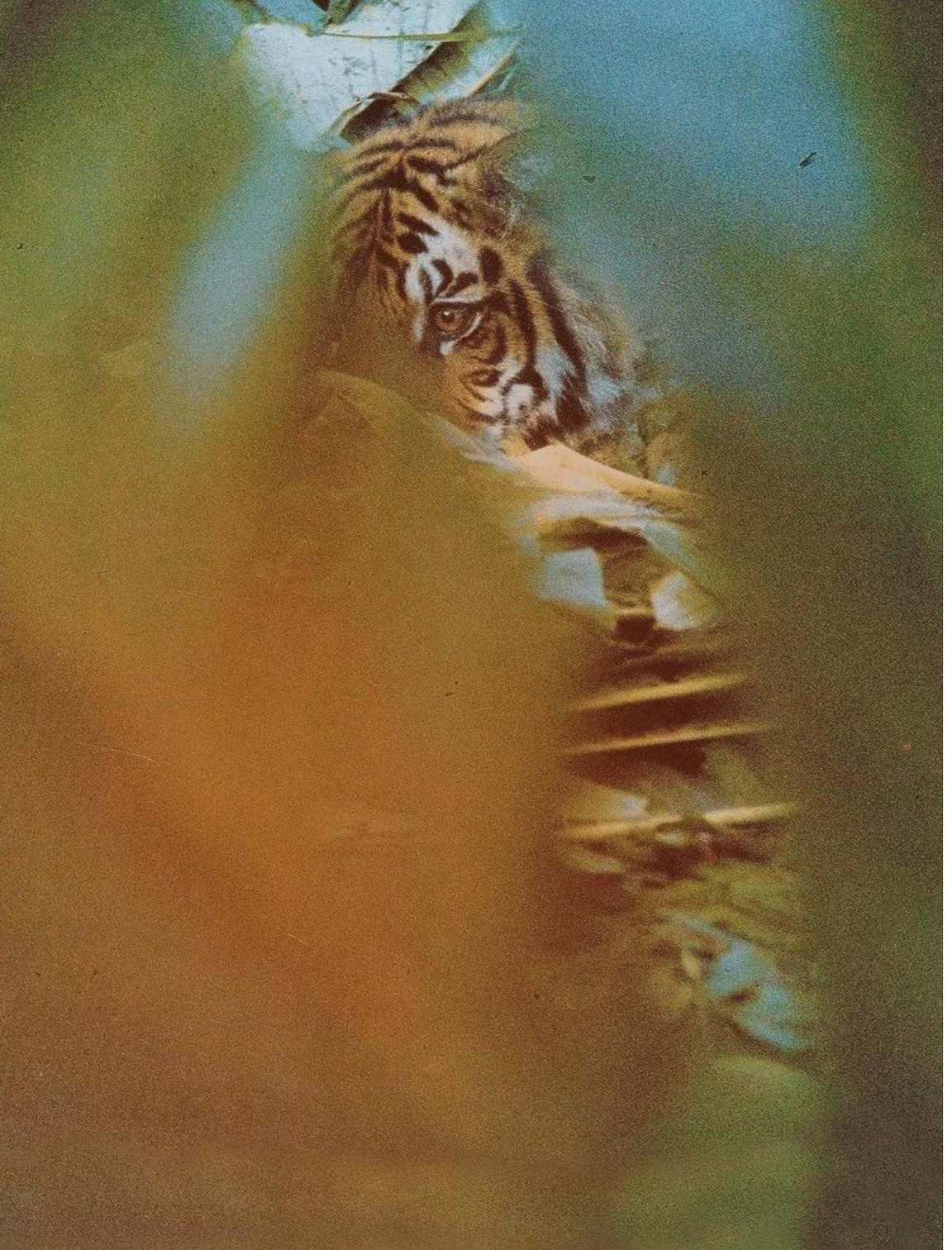
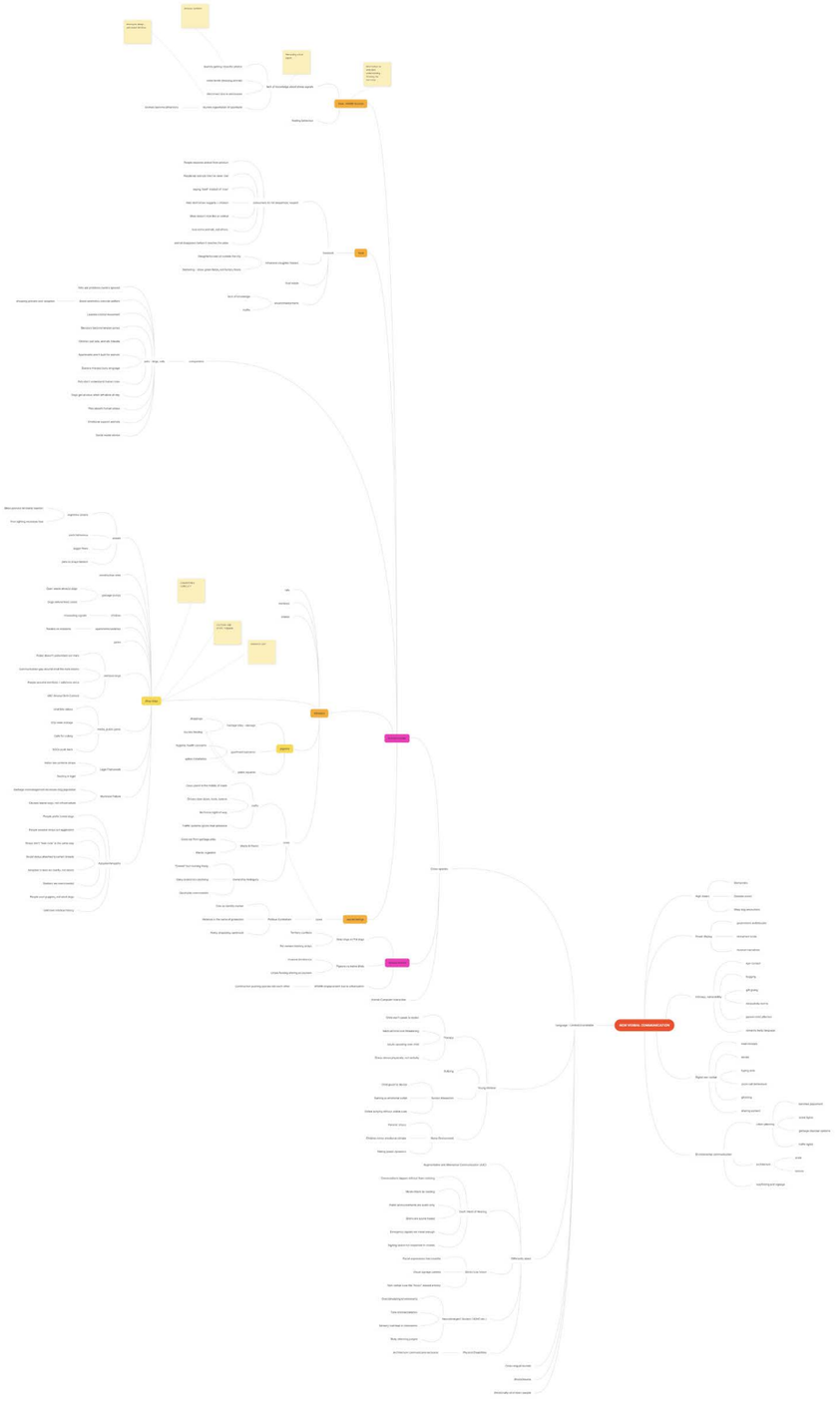
DP 1: Problem Solving

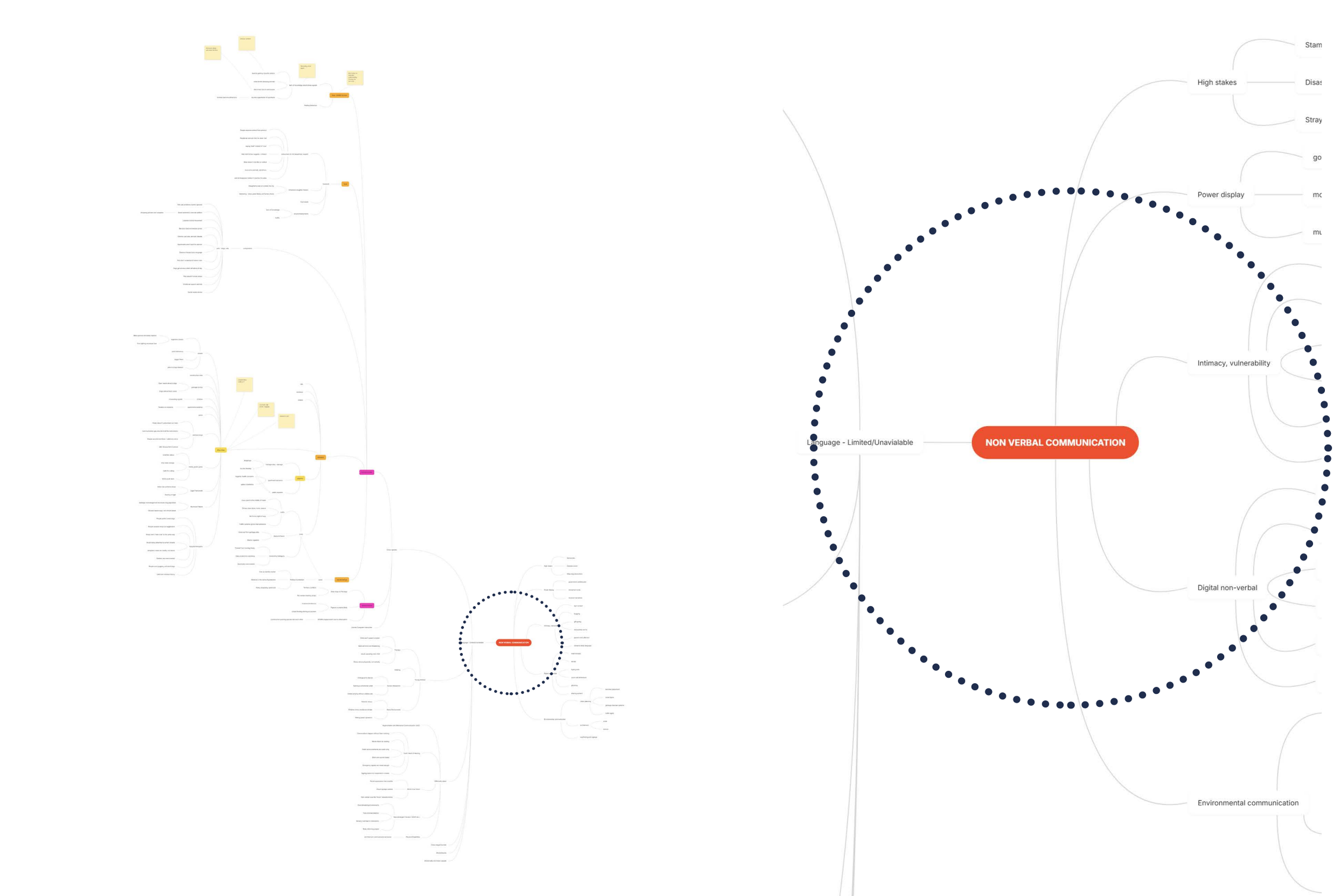


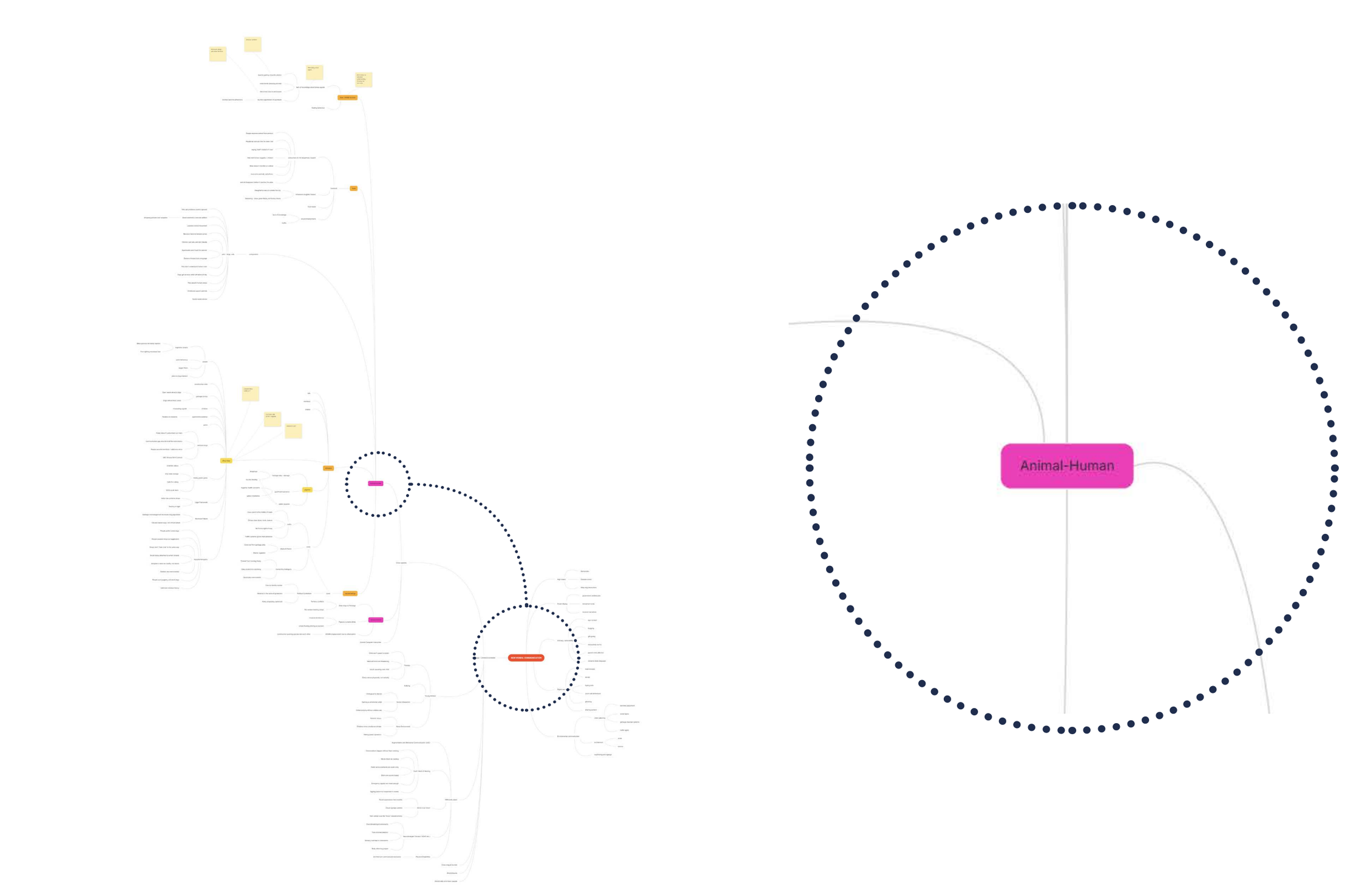
The animal is already speaking.

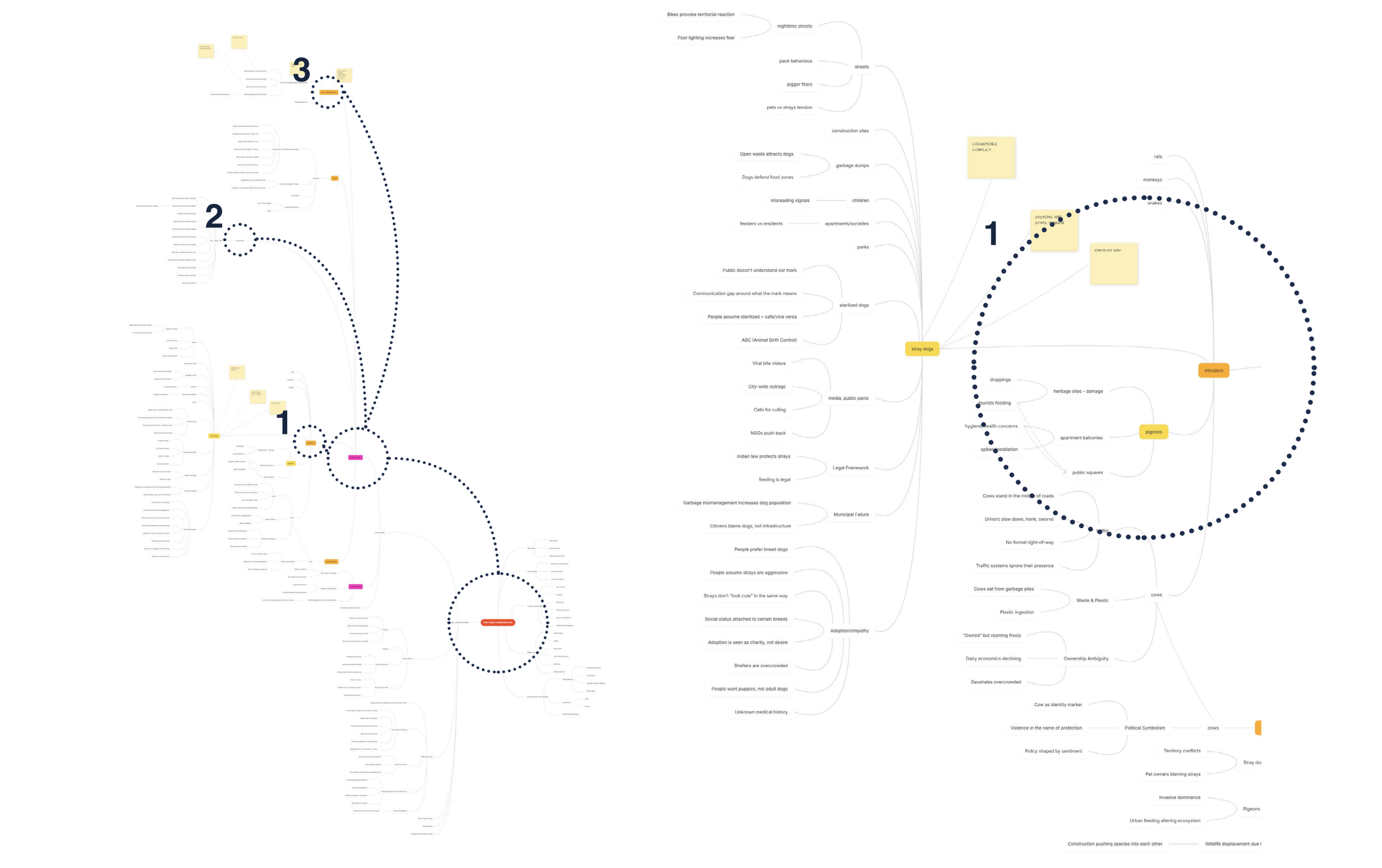
We just don’t have the equipment to receive it.

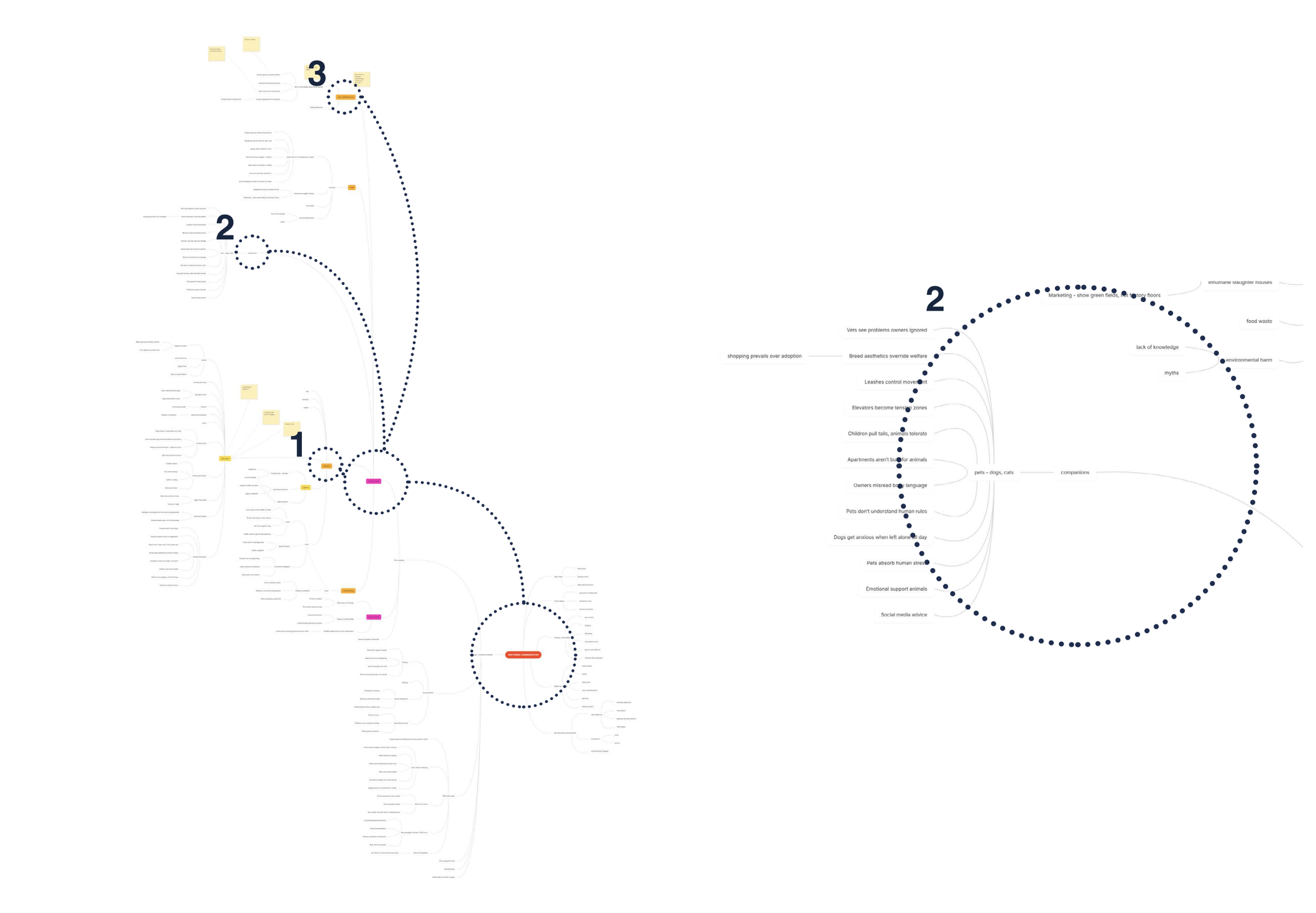


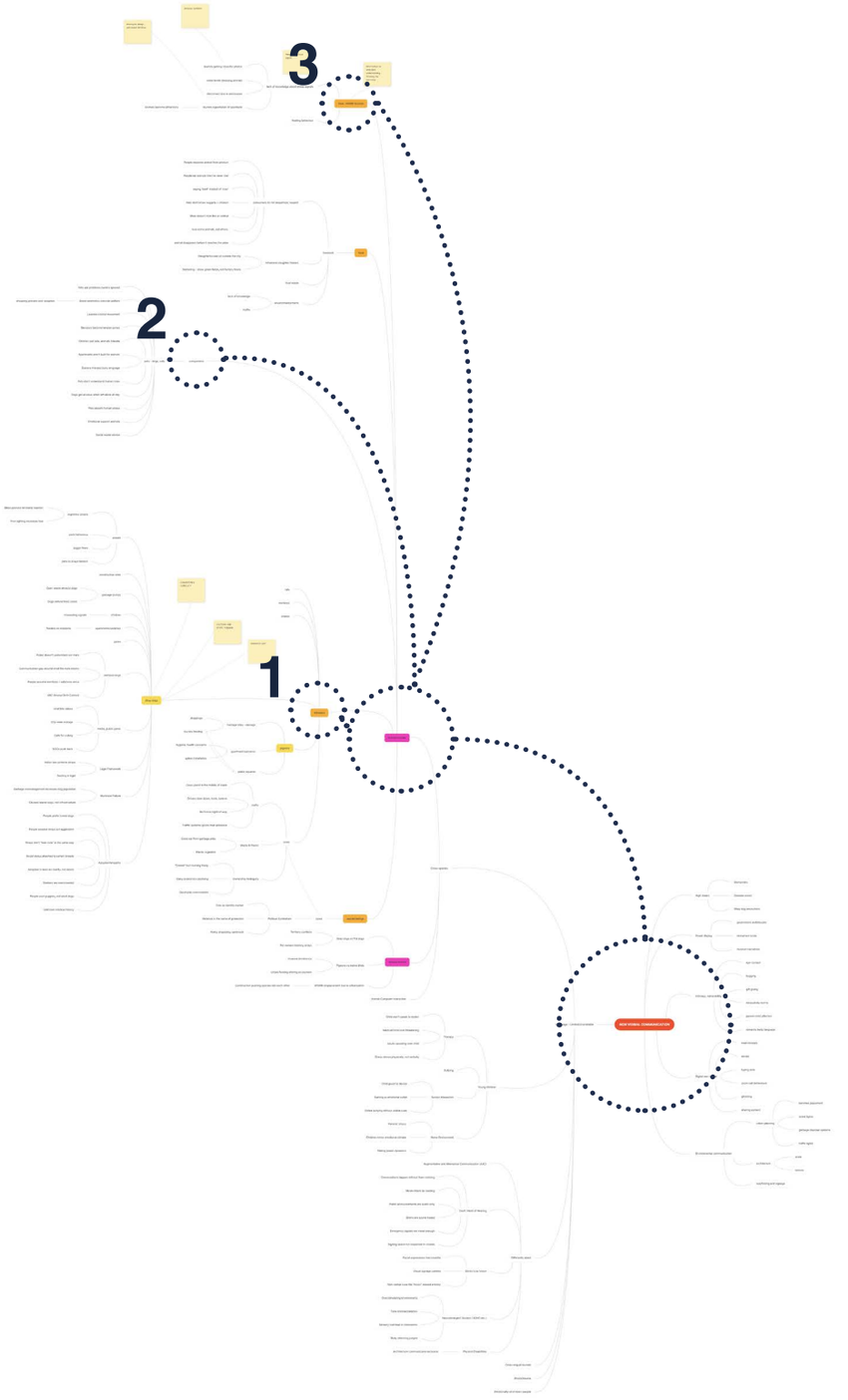
Problem Discovery

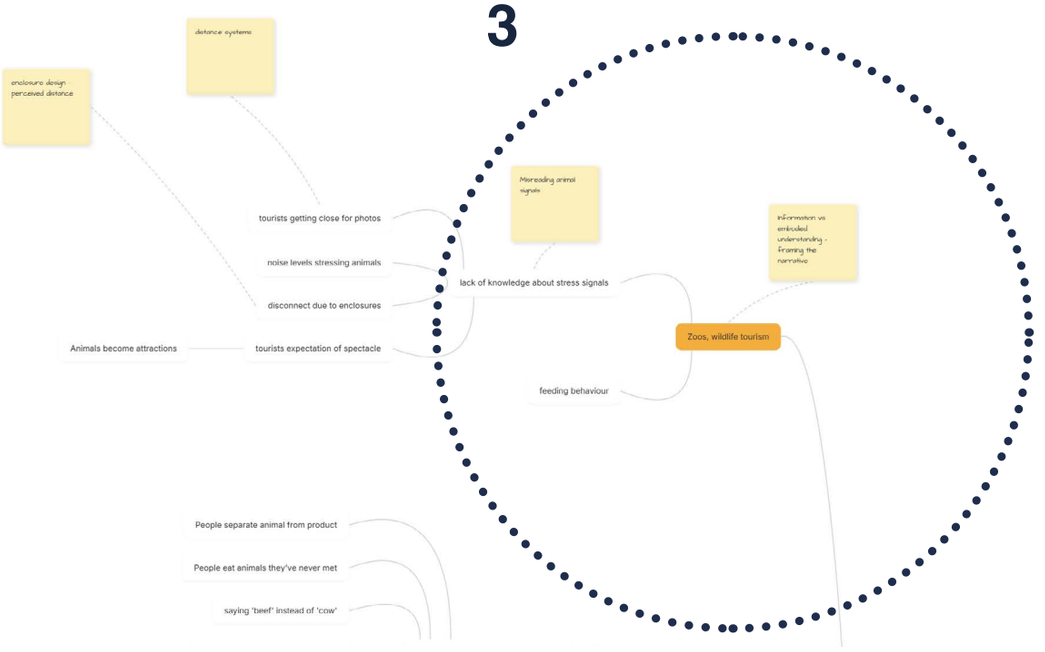

NON VERBAL COMMUNICATION
Language - Limited/Unavialable
High stakes
Power display
Intimacy, vulnerability
Digital non-verbal
Environmental communication
Stam
Disas
Stray
go
mo
mu


Animal-Human


Bikes provoke territorial reaction
nighttime streets
Poor lighting increases fear
pack behavious
streets
jogger fears
pets vs strays tension
construction sites
Open waste attracts dogs
garbage dumps
Dogs defend food zones
misreading signals
children
feeders vs residents
apartments/socieites
parks
Public doesn't understand ear mark
Communication gap around what the mark means
sterilized dogs
People assume sterilized = safe/vice versa
ABC (Animal Birth Control)
Viral bite videos
City-wide outrage
media, public panic
Calls for culling
NGOs push back
Indian law protects strays
Legal Framework
feeding is legal
Garbage mismanagement increases dog population
Municipal Failure
Citizens blame dogs, not infrastructure
People prefer breed dogs
People assume strays are aggressive
Strays don't "look cute" in the same way
Social status attached to certain breeds
Adoption/empathy
Adoption is seen as charity, not desire
Shelters are overcrowded
People want puppies, not adult dogs
Unknown medical history
stray dogs
COEXISTENCE CONFLICT
1
SYSTEMS AND STATE FAILURE
EMPATHY GAP
rats
monkeys
snakes
intruders
droppings
heritage sites - damage
tourists feeding
hygiene/health concerns
apartment balconies
pigeons
spikes installation
public squares
Cows stand in the middle of roads
Drivers slow down, honk, swerve
No formal right-of-way
Traffic
Traffic systems ignore their presence
Cows eat from garbage piles
Waste & Plastic
cows
Plastic ingestion
"Owned" but roaming freely
Dairy economics declining
Ownership Ambiguity
Gaushalas overcrowded
Cow as identity marker
Violence in the name of protection
Political Symbolism
cows
Policy shaped by sentiment
Territory conflicts
Stray do
Pet owners blaming strays
Invasive dominance
Pigeons
Urban feeding altering ecosystem
Construction pushing species into each other
Wildlife displacement due t


3
2
1
Vets see problems owners ignored
shopping prevails over adoption
Breed aesthetics override welfare
Leashes control movement
Elevators become tension zones
Children pull tails, animals tolerate
Apartments aren't built for animals
Owners misread body language
Pets don't understand human rules
Dogs get anxious when left alone all day
Pets absorb human stress
Emotional support animals
Social media advice
pets - dogs, cats
companions
Marketing - show green fields, not factory floors
inhumane slaughter houses
food waste
lack of knowledge
myths
environmental harm


3
2
1

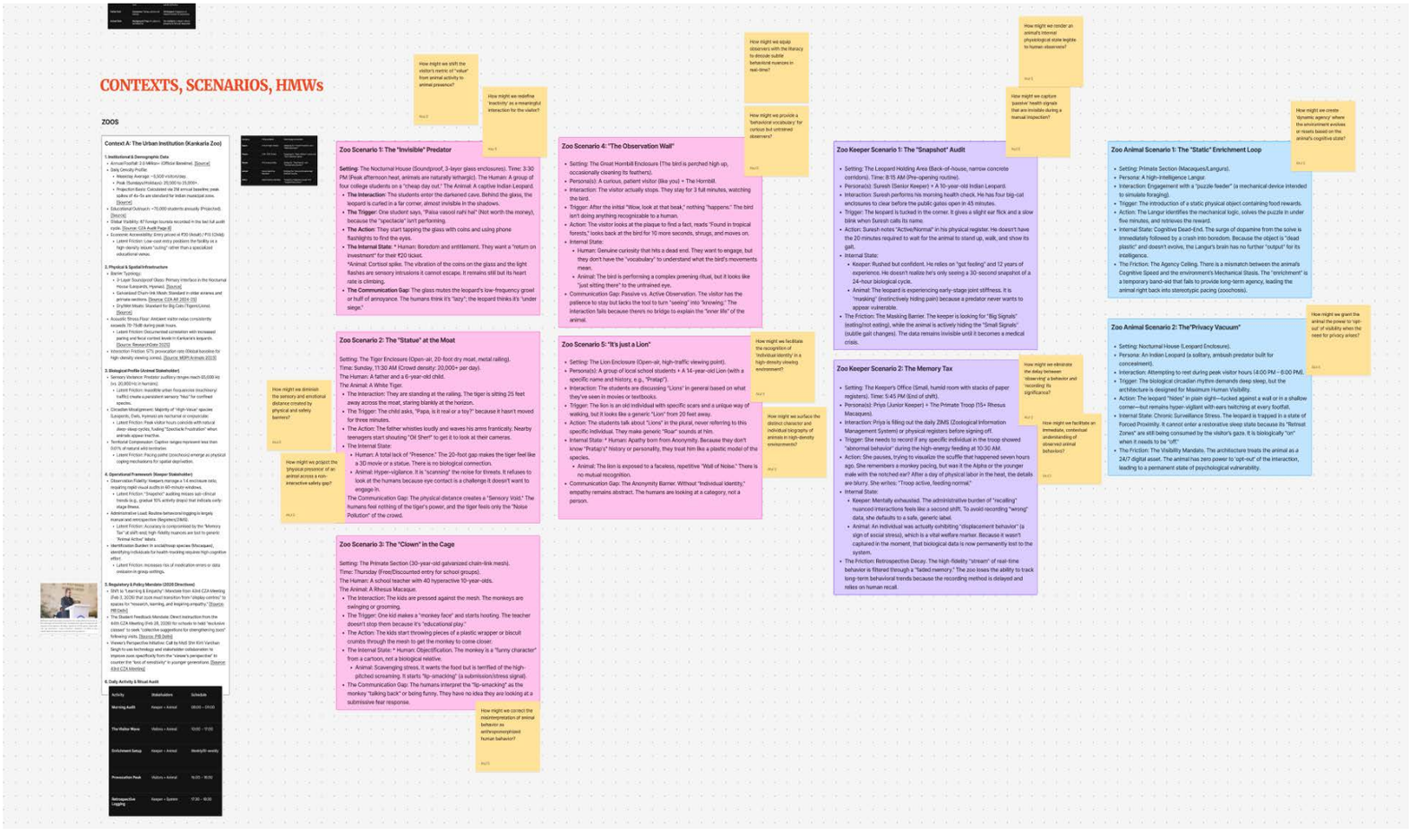

3
distance systems
enclosure design - perceived distance
Misreading animal signals
Information vs embedded understanding - framing the narrative
tourists getting close for photos
noise levels stressing animals
disconnect due to enclosures
tourists expectation of spectacle
Animals become attractions
lack of knowledge about stress signals
Zoos, wildlife tourism
feeding behaviour
People separate animal from product
People eat animals they've never met
saying "beef" instead of "cow"

WE PICKED THE ZOO



Why Zoo?

clearest place to design: highest density of encounters, most visible breakdown, and a live policy mandate to fix it.



Why Zoo?

Recognising vs. Relating

Zoo visits are mostly about recognising and identifying animals, and not relating to them. This creates a distance where visitors disengage, provoke animals to trigger a reaction, and treat them as objects of spectacle.

Taking a photo the moment they see a new animal - then moving on
Misreading behaviours through anthropomorphic labels - “it’s happy,” “it’s bored,” “it’s being lazy”
Teasing to elicit a response - and when nothing comes, apathy

Source: MDPI Animals 2023 — 57% enclosure interaction provocation rate
*Source: ResearchGate 2025 — ambient noise correlation with cortisol and pacing in captive leopard*s





At the Enclosure

Primary research:
Indroda Nature Park and Kankaria Zoo, Ahmedabad

Phones out before eyes up. Nobody read a signboard, including me.

Bigger animals got more time. Spectacle drove attention, not curiosity.

Teasing was constant and unprompted. The animal wasn't doing anything, so visitors tried to make it do something.

The leopard responded differently to different people. Calm observers got calm behaviour. Loud visitors got charges. The animal was reading the room. Nobody in the room knew that.

A keeper who had known the leopard since it was a cub called him over. The leopard snuggled with him.

A one-horned deer was moving its head up and down continuously. I wanted to know why. There was no way to find out.

The gap is not information. It is the ability to receive what the animal is already broadcasting.

The User

School-aged children. 6 to 14.
Organised group visits.
70,000 through Kankaria every year.
Highest provocation rate of any visitor group.
Lowest biodiversity literacy.
First and often only direct encounter with these animals.

What lands here lands for life.

Context A: The Urban Institution (Kankaria Zoo)

1. Institutional & Demographic Data

- Annual Footfall: 2.0 Million+ (Official Baseline). [Source]
- Daily Density Profile:
  - Weekday Average: ~5,500 visitors/day.
  - Peak (Sundays/Holidays): 20,000 to 25,000+.
  - Projection Basis: Calculated via 2M annual baseline; peak spikes of 4x–5x are standard for Indian municipal zoos. [Source]
- Educational Outreach: ~70,000 students annually (Projected). [Source]
- Global Visibility: 87 foreign tourists recorded in the last full audit cycle. [Source: CZA Audit Page 8]
- Economic Accessibility: Entry priced at ₹20 (Adult) / ₹10 (Child).
  - Latent Friction: Low-cost entry positions the facility as a high-density leisure "outing" rather than a specialized educational venue.

2. Physical & Spatial Infrastructure

- Barrier Typology:
  - 3-Layer Soundproof Glass: Primary interface in the Nocturnal House (Leopards, Hyenas). [Source]
  - Galvanized Chain-link Mesh: Standard in older aviaries and primate sections. [Source: CZA AR 2024-25]
  - Dry/Wet Moats: Standard for Big Cats (Tigers/Lions). [Source]
- Acoustic Stress Floor: Ambient visitor noise consistently exceeds 70-75dB during peak hours.
  - Latent Friction: Documented correlation with increased pacing and fecal cortisol levels in Kankaria's leopards. [Source: ResearchGate 2025]
- Interaction Friction: 57% provocation rate (Global baseline for high-density viewing zones). [Source: MDPI Animals 2023]

3. Biological Profile (Animal Stakeholder)

- Sensory Variance: Predator auditory ranges reach 65,000 Hz (vs. 20,000 Hz in humans).
  - Latent Friction: Inaudible urban frequencies (machinery/traffic) create a persistent sensory "hiss" for confined species.
- Circadian Misalignment: Majority of "High-Value" species (Leopards, Owls, Hyenas) are nocturnal or crepuscular.
  - Latent Friction: Peak visitor hours coincide with natural deep-sleep cycles, fueling "Spectacle Frustration" when animals appear inactive.
- Territorial Compression: Captive ranges represent less than 0.01% of natural wild territories.
  - Latent Friction: Pacing paths (zoochosis) emerge as physical coping mechanisms for spatial deprivation.

4. Operational Framework (Keeper Stakeholder)

- Observation Fidelity: Keepers manage a 1:4 enclosure ratio, requiring rapid visual audits in 60-minute windows.
  - Latent Friction: "Snapshot" auditing misses sub-clinical trends (e.g., gradual 10% activity drops) that indicate early-stage illness.
- Administrative Load: Routine behavioral logging is largely manual and retrospective (Registers/ZIMS).
  - Latent Friction: Accuracy is compromised by the "Memory Tax" at shift-end; high-fidelity nuances are lost to generic "Animal Active" labels.
- Identification Burden: In social/troop species (Macaques), identifying individuals for health-tracking requires high cognitive effort.
  - Latent Friction: Increases risk of medication errors or data omission in group settings.

5. Regulatory & Policy Mandate (2026 Directives)

- Shift to "Learning & Empathy": Mandate from 43rd CZA Meeting (Feb 3, 2026) that zoos must transition from "display centres" to spaces for "research, learning, and inspiring empathy." [Source: PIB Delhi]
- The Student Feedback Mandate: Direct instruction from the 44th CZA Meeting (Feb 28, 2026) for schools to hold "exclusive classes" to seek "collective suggestions for strengthening zoos" following visits. [Source: PIB Delhi]
- Viewer's Perspective Initiative: Call by MoS Shri Kirti Vardhan Singh to use technology and stakeholder collaboration to improve zoos specifically from the "viewer's perspective" to counter the "loss of sensitivity" in younger generations. [Source: 43rd CZA Meeting]

CONTEXT

Kankaria Zoo, Ahmedabad

2M+ annual visitors.
Peak density 20,000/day.
Ambient noise 70–75dB, documented correlation with increased cortisol and pacing in leopards.
Enclosure interaction provocation rate: 57%.

CONTEXT

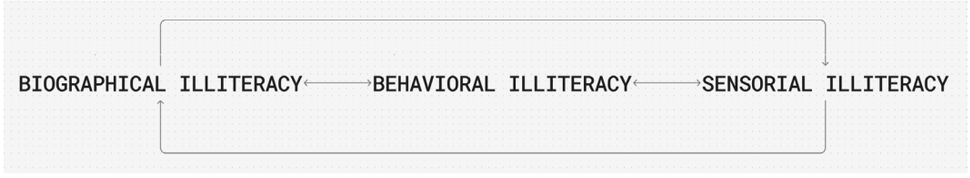
Regulatory & Policy Mandate (2026 Directives)



Referring to rapid technological advancement, Shri Singh called for discussions on how technology can be used for better zoo management, improved animal care and enhanced visitor experience. He further stressed the need for greater collaboration with state governments, research institutions, stakeholders and NGOs to gain valuable inputs for improving zoos from the viewer's perspective.

- Shift to "Learning & Empathy": Mandate from 43rd CZA Meeting (Feb 3, 2026) that zoos must transition from "display centres" to spaces for "research, learning, and inspiring empathy." [Source: PIB Delhi]
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Three Illiteracies. One Loop.



Biographical illiteracy — I don't know who you are as an individual
Behavioral illiteracy — I can't read what your body is saying
Sensorial illiteracy — I cannot perceive the frequency you exist in

Signs, guides, and plaques address biographical illiteracy at best. Behavioral and sensorial remain mostly unattended to.

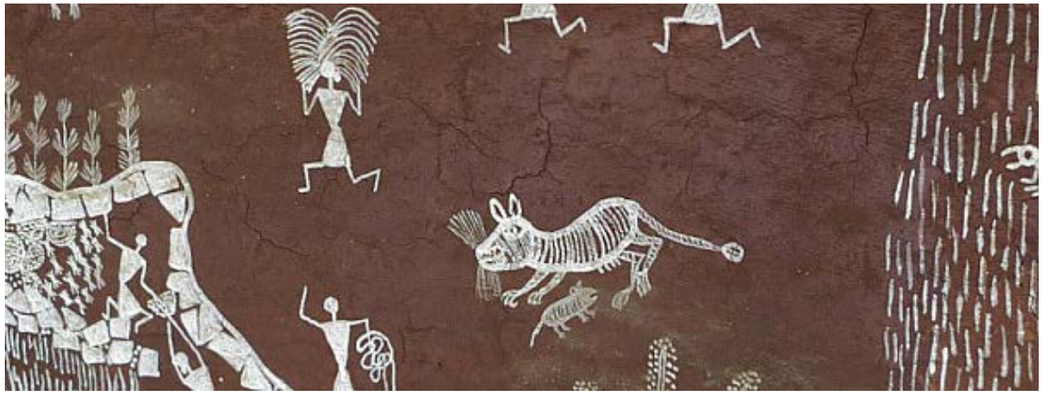
Zoo visitors score ~3/10 on biodiversity literacy. They consistently misread stress responses as entertainment. (Global Evaluation of Biodiversity Literacy)
Stillness ≠ laziness. It is vigilant mapping. (Felid Sensory Biology)
Leopards operate in infrasound and spatial acoustic maps imperceptible to humans. The gap is not social — it is hardware. (Umwelt, von Uexküll)
43rd CZA Meeting, 2026: mandate for Indian zoos to shift from "display centres" to spaces for "research, learning, and inspiring empathy."

This knowledge existed. We designed it out.



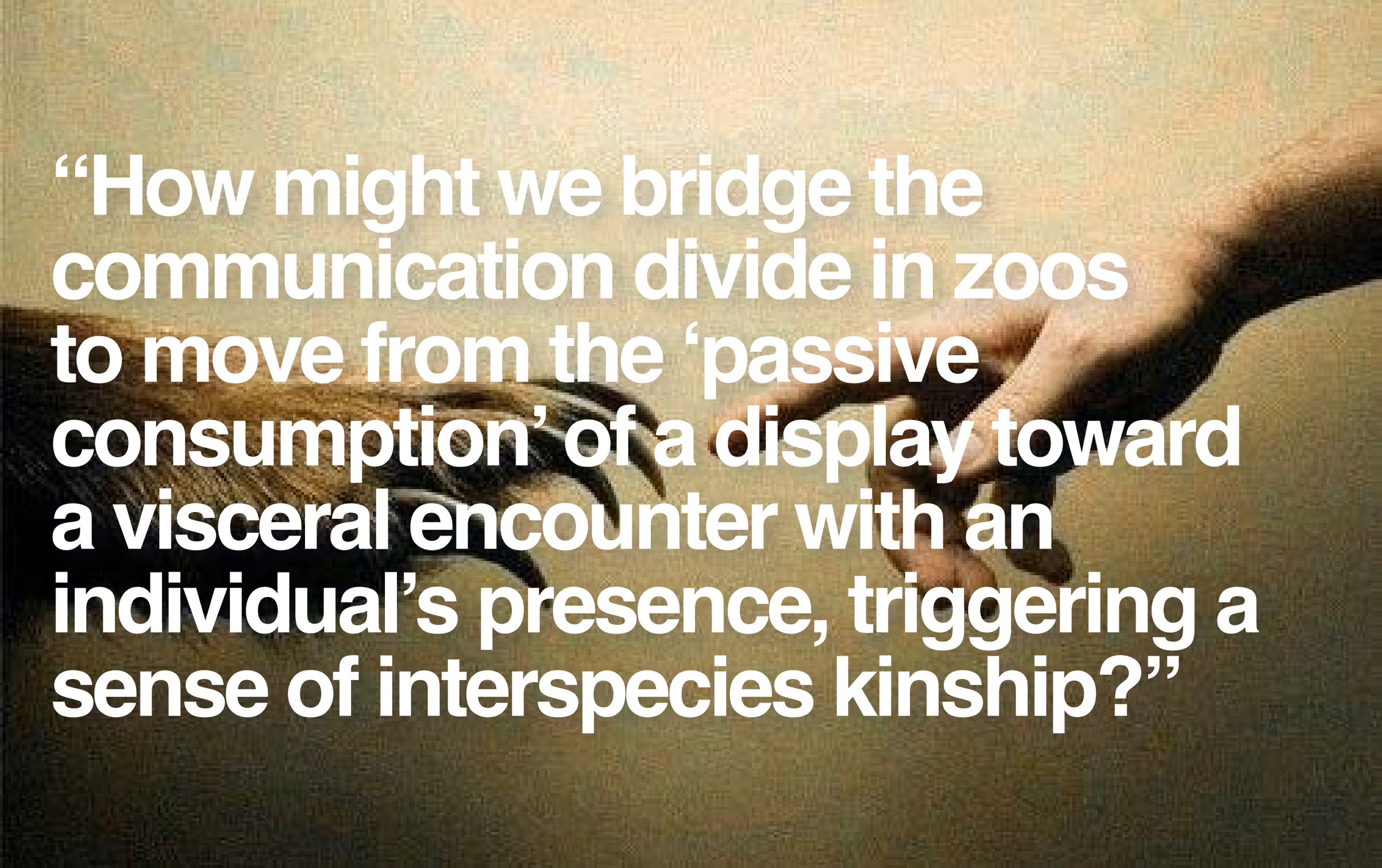
Indigenous and traditional communities did not have this gap.

- **Warli tradition**: humans and animals depicted at equal scale, in shared ecological space. Not hierarchy. Cohabitation.
- **Waghoba**: Western Ghats leopard/tiger deity. The animal is not an exhibit, but a presence to be respected. Territorial boundaries acknowledged before approach.
- **Shamanic soul-transfer**: cross-cultural practice of temporarily inhabiting an animal's body schema to understand its state. Embodied empathy as method, not metaphor.



From Spectator, to Kin:

Dimension	The Spectator	The Kin
Level of Perception	"It's a Lion."	"It's an individual with a history/mood."
Driver of Curiosity	Spectacle: waiting for a roar	Discovery: noticing subtle patterns
Visitor Role	Consumer: photo and leave	Participant: shared awareness
Animal Role	Background prop	Co-resident: presence felt + respected


“How might we bridge the communication divide in zoos to move from the ‘passive consumption’ of a display toward a visceral encounter with an individual’s presence, triggering a sense of interspecies kinship?”



The Tiger's Umwelt

- **Infrasound Reception:** territorial communication felt in the chest, below human hearing
- **Pinna Directionality:** involuntary ear rotation to map threats
- **Stress Heartbeat:** 40–50 BPM at rest, spikes above 100 during crowd frustration
- **Flehmen Response:** drawing scent into a second nose to read social status
- **Stereotypic Pacing:** frustration from spatial deprivation made physical



The Tiger's Umwelt

What you see	What is actually happening
"It's just lying there. Boring."	40–50 BPM resting heart rate. Ears rotating independently, mapping every footfall in the corridor. Vigilant acoustic mapping — not sleep.
"It's ignoring us."	Infrasound reception — processing territorial signals below 20Hz, completely inaudible to humans. It is not ignoring. It is listening to a frequency we cannot hear.
"It's pacing, it must be excited."	Stereotypic pacing — zoochosis. Repetitive locomotion from spatial frustration. A coping mechanism, not excitement. The zoo's own failure, made physical.
"It twitched its ear at me."	Pinna directionality — involuntary rotation to locate and triangulate a threat source. The ear moved because a child shouted three enclosures away.
"It's sniffing the air, cute."	Flehmen response — drawing scent into the Jacobson's organ to read the reproductive and social status of every animal and human in the space.


SPANDA

SPANDA

Sanskrit: *the first pulse.*
The vibration that precedes
understanding.



SPANDA

Held against the chest.
When you hold it, you receive the animal's state, as a sensation, not information.

To receive another's presence, you must first choose to be present. Deliberate, conscious.


SPANDA


SPANDA


SPANDA


SPANDA



How haptics carry feeling

Most educational media routes through the brain first; signs, screens, audio guides land in the cognitive layer, the same layer where we project narratives onto what we see.

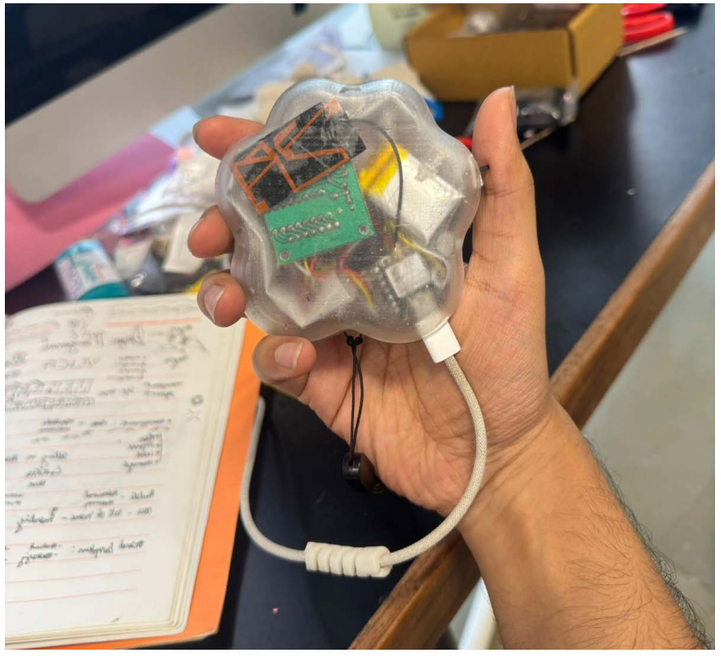
Haptics bypass that layer entirely.

Two approaches:

Where a biological rhythm exists, such as a heartbeat or a breath, we mirror it directly.
Where it doesn’t, we use rhythm rate and envelope shape as the emotional lever. Slow and smooth reads as calm. Fast and jagged reads as threat. The body already knows this.

- *Rhythm is the single parameter that independently modulates valence, arousal, and perceived roughness. (Seifi & MacLean, 2013)*
- *Only naturalistic, biomimetic patterns reliably reach calm-pleasant states in children. (Seifi & MacLean, 2013; Shim & Tan, 2020)*





The Object

Paw-shaped.
Handheld.



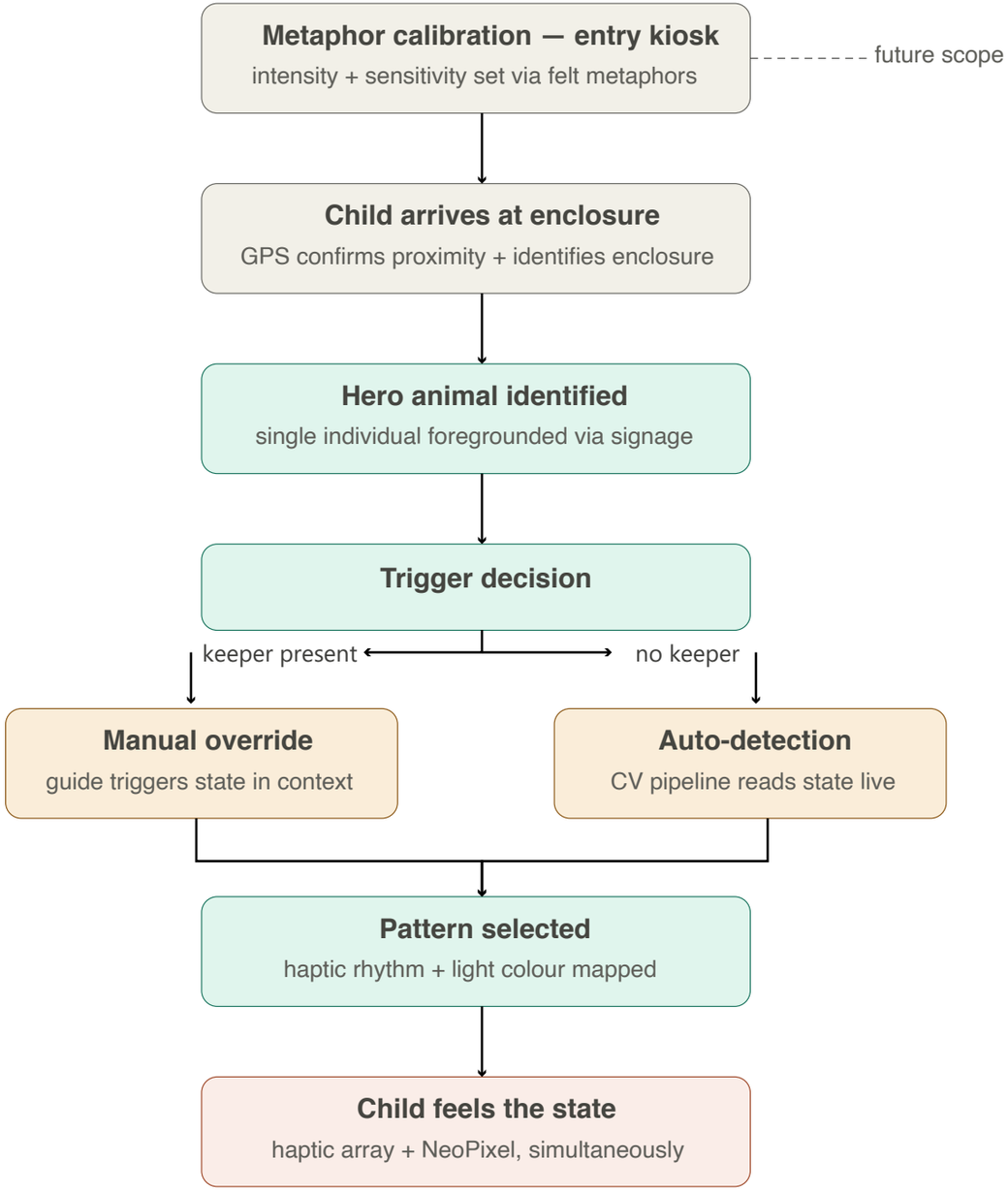
The Object

Shell translucent tinted polycarbonate, satin finish. Electronics visible inside.
Pads soft-touch silicone rubber. 5 contact points, 5 actuators.
Lanyard braided nylon. Secure handling in high-density environments.

Inside
DA7280 LRA central pad, heartbeat + resonance.
4× ERM toe pads, directional rhythm.
NeoPixel strip edge, valence layer.
ESP32-C3 wireless control.
Neo6M GPS locator
TTP223 touch sensor for activation
Translucent by design. The child sees it working. Transparency as consent.

How it works

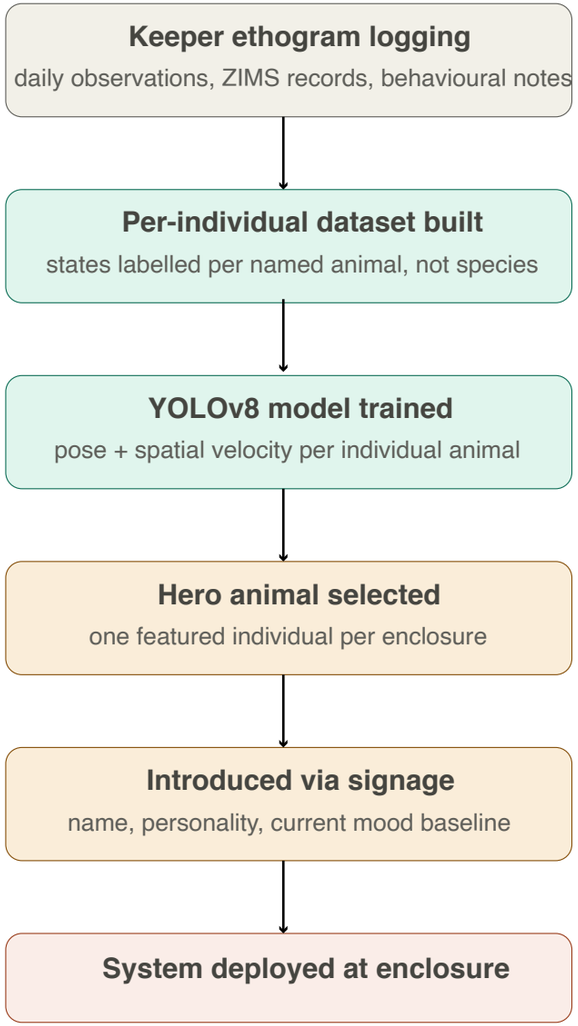
Live Experience Flow



The child calibrates at entry, during a brief onboarding, to felt metaphors. Intensity set, mystery intact. At the enclosure, GPS confirms which animal they are standing in front of. In multi-animal enclosures, one hero individual is foregrounded via signage. The trigger is either a keeper who knows the animal's current state, or the CV pipeline reading it live. Either way, what arrives in the child's chest is not information, it is presence.

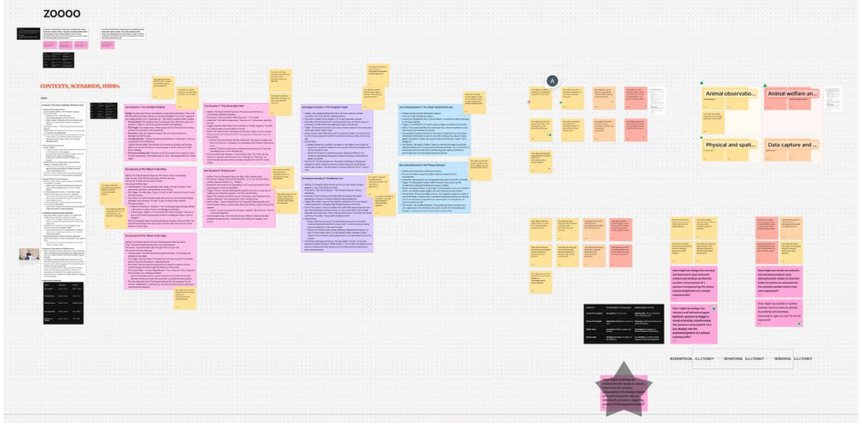
How it works

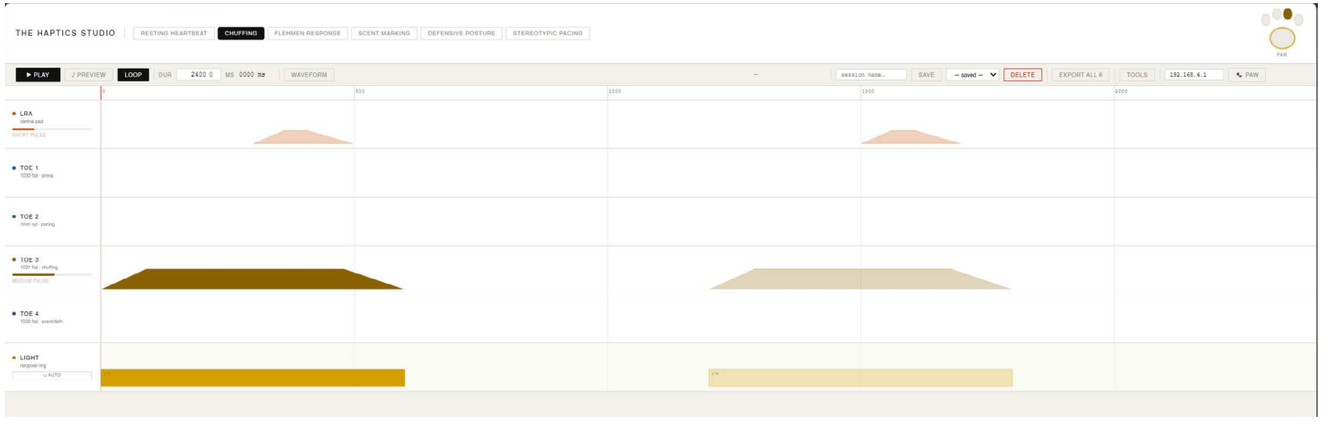
Training Pipeline

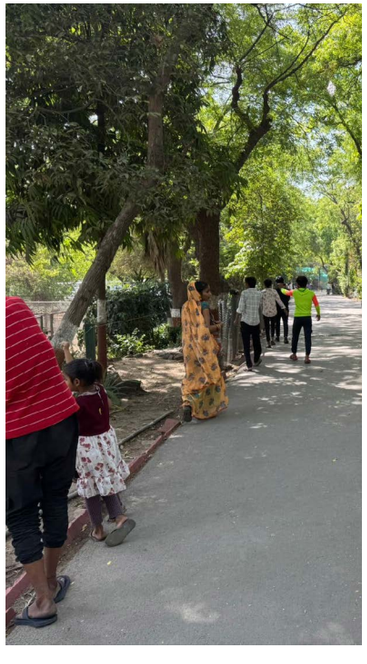


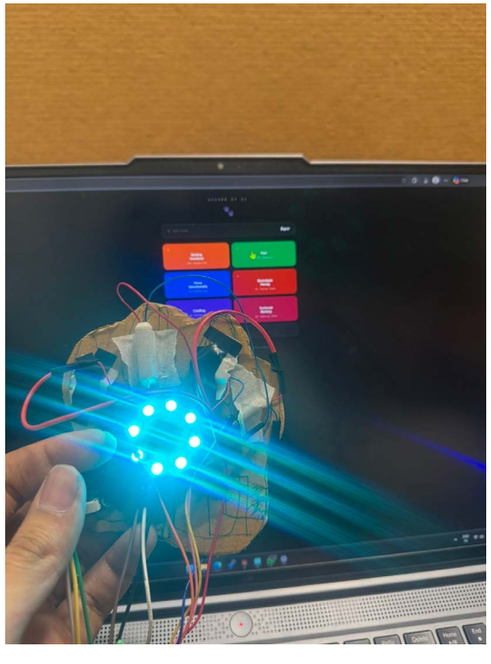
SPANDA is not trained on speciesm it is trained on individuals. Keepers log daily behavioural observations per named animal. That data builds the model. The hero animal selected for each enclosure is the one the child will feel. ‘Prathap’, not just some leopard. This is what makes attribution possible, and attribution is what makes empathy possible.

Process

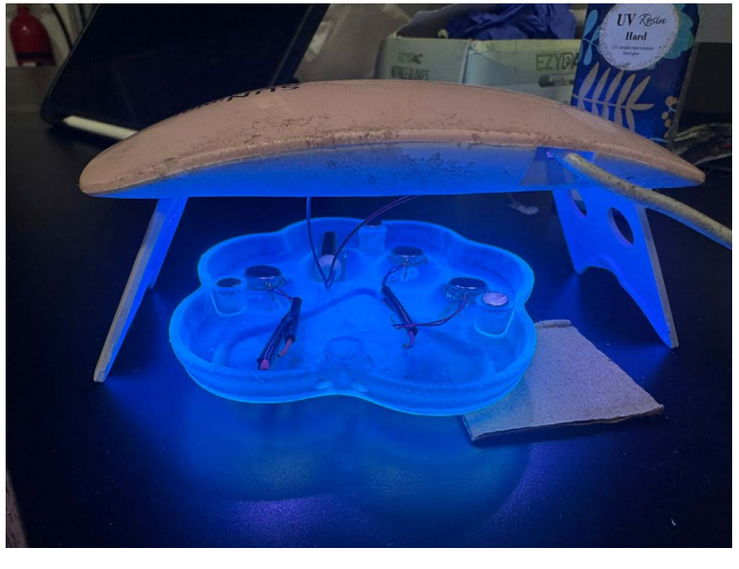








Category	Trait/Perception	Biological Cause / Internal State	Potential Technical Simulation (Actuator/Sensor)
Vision	Night/Low-Light Clarity	Tapetum Lucidum: A mirror-like layer behind the retina reflects light back for a second chance at absorption.	Visual Filter: Real-time digital brightening (ISO gain) or light-amplification overlay in a viewfinder/HUD.
	Motion-Priority Focus	High rod-to-cone ratio: rods detect motion/shape, while fewer cones mean less color saturation.	Visual Filter: High-contrast "edge detection" that highlights moving objects while desaturating static backgrounds.
	Binocular Overlap	Forward-facing eyes create a 3D image for accurate depth/distance assessment during a stalk.	Hardware: Binocular/VR lenses with a narrow, focused field of view to simulate "tunnel vision" on a target.
Hearing	Infrasound Reception	Sensitivity to low frequencies (below 20Hz) used for long-distance communication (territorial/social).	Haptic/Sub-Bass: Sub-woofer/bone-conduction at frequencies you feel in your chest/skull rather than hear in your ears.
	Frequency Sensitivity	Most sensitive in the 0.3 to 8.5 kHz band; cochlear tuning comparable to humans but sharper.	Audio Filter: High-pass filter that suppresses high-frequency human noise and amplifies low-frequency animal growls/huffs.
	Pinna Directionality	Involuntary ear-flicking/rotation to map auditory threats.	Haptic: Directional vibration on a headband or device handles that "pulls" your attention toward a sound source.
Olfactory	Flehmen Response	Drawing scents/pheromones into the Vomeronasal organ (Jacobson's organ) to identify social or reproductive status.	Olfactory Display: A wearable "Electronic Nose" (e-nose) that releases specific scents (e.g., musk) when it detects bio-chemicals.
	Scent Mapping	Territorial messages left via urine or paw-pad glands.	Visual Overlay: AR "heat-trail" or "scent-glow" that visualizes chemical markers left on trees/enclosure walls.
Tactile	Paw Sensitivity	Large, padded paws for soundless movement (stealth).	Kinesthetic: High-density foam or soft-touch material on the device grip that simulates the "give" of a padded paw.
Physiology	Stress Heartbeat	40-50 bpm in deep rest, spikes (100+) during frustration/stress.	Haptic Thumper: A voice-coil actuator pulsing in the user's palm at a frequency synced to the animal's state.
	Panting/Thermal Load	Mouth open breathing used for evaporative cooling (thermoregulation)	Thermal: Peltier plate that fluctuates in temperature getting hot during "stress" and cooling down during "rest."
	Piloerection	Involuntary "hackles" raising along the spine during fear or high arousal.	Tactile: A fabric or surface on the device that physically "stiffens" or changes texture (using electrostatic or mechanical means).
	Tail Flicking / Twitching	Rising arousal, focus, or irritation; internal engine "starting."	Haptic: Sharp, whip-like jitter or "flick" pulse at the fingertips/grip handles.
Cognitive Agency	Hunting / Stalking Intent	Calculating distance and timing; the mental "lock" on a target.	Focus HUD: A visual tightening of the field of view + a haptic "tension" that releases only when the animal breaks eye contact.
	Environmental Mapping	A 3D mental map of every rock, tree, and hiding spot in the enclosure.	Wayfinding: A "ghost sense" that pulls the user's attention to high-vantage points or "safe zones."
	Problem Solving / Play	Interaction with enrichment or novel objects as an expression of curiosity.	Cognitive Resistance: A physical "push back" from the device (e.g., the "Tiger vs. Log" weight simulation).
	Territorial Confidence	The sense of "owning" a space via scent and presence.	Spatial Aura: A subtle "background thrum" that gets stronger/steadier in areas the cat has heavily marked.
Emotional Agency	Coping / Resilience	Choosing to retreat to a quiet spot to handle the stress of 2 million annual visitors.	Isolation Buffer: The device goes "silent" or "cold" when the cat moves to a private area.
	Avoidance Behavior	Deliberate choice to seek privacy; a manifestation of solitary agency.	Simulation: Device becomes non-responsive or "dimmed" to mirror the animal's withdrawal.
	Defensive Posture	Pinned ears, hissing, or growling in response to perceived threats/crowd noise.	Simulation: Jagged, high-arousal vibration + directional audio huffs/hiss via bone conduction.
	Affective States	Complex moods ranging from "Agitated" to "Playful" or "Melancholic."	Mood Lighting/Tone: A subtle change in the device's color or the "Rotom-dex" guide's tone.
	Individual Temperament	Distinct personality traits (e.g., bold vs. shy); baseline arousal.	Biographical Layer: The system identifies the specific resident and adjusts simulation baselines to match.
	Stereotypic Pacing	Frustration from lack of habitat complexity or anticipation of food.	Haptic: Rhythmic, heavy-impact "steps" felt through the device handles to simulate heavy pacing.
Social & Relational	Social Lean (Proximity)	Choosing to sleep or rest in physical contact with a specific individual. The most frequent sign of social affiliation.	Constant, slow warmth + shared pulse.
	Roaring / Moaning	Territorial marking and calling to "near and dear ones."	Haptic/Sub Bass: Intense sub bass vibration through the chest-plate + directional audio.
	Contact Grunts / Chuffing	Social checking: "I am here, are you okay?" / Non-threatening greeting.	Haptic: Rhythmic, low frequency internal thrum or audio "huff" via bone conduction.
	Head Rubbing / Nuzzling	Affirming social bonds and interspecies kinship.	Thermal: Broad thermal warmth + slow, steady kneading pressure on the device grips.
	Grooming	Strengthening trust and reducing social tension after a crowded day.	Tactile: Rhythmic "scraping" haptic sensation + localized warmth.
Biological Maintenance	Self-Actualization	The "satisfaction" of a perfect sun-soak or a successful groom.	Thermal Contentment: A slow, radiating heat and low-frequency "purr" (low-frequency rumble).



Feel it

DEMO

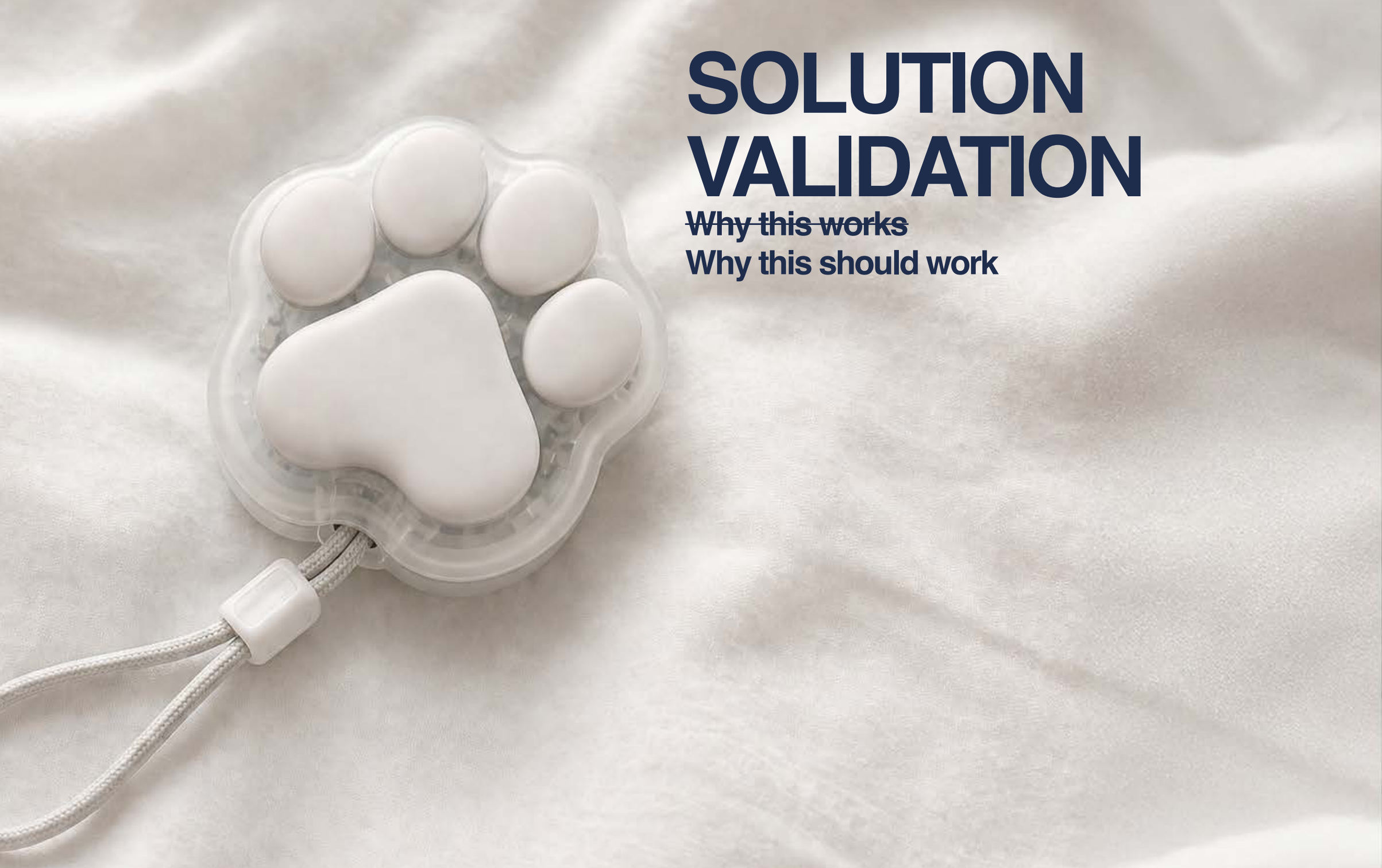


Three states.
Resting heartbeat, Stereotypic Pacing, Chuffing.
Feel rest. Feel stress. Feel the sniff.



SOLUTION VALIDATION

Why this works



SOLUTION VALIDATION

~~Why this works~~

Why this should work



SOLUTION VALIDATION

~~Why this works~~

Why this should work, hopefully*

Philosophical — von Uexküll, 1934 — The gap is biological, not social.
Body site — McGlone et al., 2014; Löken et al., 2009 — Chest CT afferents read signal as internal.
Haptic rhythm — Seifi & MacLean, 2013 — Rhythm is the primary emotional lever.
Haptic patterns — Shim & Tan, 2020 — Biomimetic patterns reach calm states; abstract codes fail.
Pattern accuracy — Jones et al., 2009 — Torso yields 99–100% pattern recognition accuracy.
Valence — Ju & Minamizawa, CHI EA 2023 — Vibration conveys valence without projection.
Psychological — Conservation psychology — Empathy predicts conservation behaviour.
Precedent — Chen et al., IJHCI 2025 — Haptic wearables increase nature connectedness in zoos.



VISION

What this becomes

- **Onboarding and calibration** The metaphor-based calibration exists as a concept. The kiosk interface, the child-facing flow, the sensitivity logic are all unbuilt. Designing that experience is the next design problem.
- **The keeper interface** Currently a bare web panel for demo triggering. A proper keeper interface would let staff log behavioural observations in real time, flag state changes, and build the ethogram directly through daily interaction with the device. A full design project in itself.
- **Signage redesigned around biography** Not “Indian Leopard, Panthera pardus fusca.” Dharma. 7 years old. Currently resting. Signage that makes the animal a subject, not a specimen.
- **A haptic vocabulary that learns** Each calibration a data point. The system gets better at knowing which patterns land for which bodies. The framework keeps getting learnt.
- **A translation framework** One that doesn’t exist yet in this project or in the literature. SPANDA is the first attempt to need one.
- **Richer sensor input** CV pipeline currently reads pose and spatial velocity. Future: directional microphones picking up vocalisation, thermal cameras reading body heat and respiration, ground vibration sensors detecting movement through substrate. More input streams mean more states readable, more of the animal’s inner life translatable. Including when it isn’t visible.

SPANDA. The first pulse.

FAQs
(frequently asked to me, by me)

- **Why children specifically?**
  Highest provocation rate. Lowest literacy. Longest memory. A felt experience at 8 years old shapes how you see an animal for the rest of your life.
- **Why the chest and not the wrist?**
  The chest contains CT afferents that route to the interoceptive cortex. The body reads the signal as coming from within, not from a device. The wrist is informational touch. The chest is affective touch. (McGlone et al., 2014)
- **Why haptics and not audio?**
  Sound is processed cognitively. We assign meaning, narrative, and emotion to what we hear. Haptics bypass that layer. The body receives before the brain interprets.
- **What if the animal isn't visible?**
  SPANDA defaults to resting heartbeat. Presence felt even when presence is hidden.
- **What about multi-animal enclosures?**
  One hero individual foregrounded per enclosure, introduced by name via signage. SPANDA is calibrated to that individual.
- **What if there's no keeper?**
  CV pipeline handles it. Keeper presence enables richer contextual triggering. No keeper means the system runs autonomously. Both are valid.
- **How do you handle haptic subjectivity?**
  Metaphor-based calibration at entry sets individual intensity and sensitivity. Long term, each calibration becomes a data point for a learning model.
- **How does this scale?**
  Keepers already log daily behavioural observations. SPANDA plugs into that existing workflow. Each enclosure gets a trained model per individual animal. The pipeline is repeatable.


Thank you!